

Research is an art as well as a science.

Research Methods Vs Methodology:

Research methods include all those techniques that are adopted for conducting research.

Research methodology is the way in which research problems are solved systematically. It is a science of doing research.

It is not only important for the researcher to know the research methods (techniques), but also the scientific approach called methodology.

Steps in Research Process

Research process contains a series of closely related activities to carry out research.

1. Selection of Research Topic/ Area
2. Extensive Literature Survey
3. Define Research Problem
4. Formulation of Research Question, Objectives and Hypothesis
5. Preparing the Research Design
6. Define Population and Sampling
7. Data collection
8. Data Analysis
9. Hypothesis Testing
10. Generalization and Interpretation
10. Preparation of Report

Selection of Research Topic or Problem

The research topic or problem should be practical, relatively important, feasible, ethically and politically acceptable. It is selected based on preliminary studies

Literature Review or Extensive Literature Survey

After the selection of research problem, the second step is that of literature mostly connected with the topics. The availability of the literature may bring ease in the research. For this purpose academic journals, conference and govt. reports and library must be studied.

Define Research Problem

- **What it is:** A specific issue, gap, or challenge within the topic that needs investigation.
- **Purpose:** Justifies why the research is necessary.
- **Example:** *Retail HR departments struggle to adapt to digital tools, leading to inefficiencies.*

Formulation of Research Questions, Objectives and Making Hypothesis: Define what is to be discovered or tested. Hypothesis means assumptions, which may be true, partially true, or wrong. The hypothesis is to draw the positive and

negative cause and effect aspects of a problem. Hypothesis narrows down the area of a research and keep a researcher on the right path.

Preparing the Research Design

Choose methodology (qualitative, quantitative, or mixed). Decide on tools, techniques, and procedures for data collection. Detail plan on how to gather data for testing of hypothesis

Research design may be made, depend on the nature and purpose of the study. During Research Design the information about sources, skill, time and finance is taken into consideration.

Define Population and Sampling

The researcher must select population and design a sample from where he/she can get data. The sample may be of two types:

1. Probability Sampling
2. Non-probability Sampling

Data collection

Data collection is the most important work, is researcher. The collection of information must be containing on facts which is from the following two types.

Primary Data Collection: Primary data may be from the following.

1. Experiment
2. Questionnaire
3. Observation
4. Interview

Secondary data collection: it has the following categories:

1. Review of literature
2. Official and non-official reports
3. Library approach

Data Analysis

To draw inference.

Data analysis may be divided into two main categories.

- Data Processing -- Data editing, Data classification, Data tabulation, Data presentation, Data measurement
- Data Exposition -- Description, Explanation, Narration, Conclusion/Findings, Recommendations/Suggestions

Hypothesis Testing

Based on data analysis hypothesis is tested whether to accept or reject the hypothesis.

Generalization and Interpretation

The acceptable hypothesis is used to arrive at the process of generalization or to make a theory. Some types of research have no hypothesis for which researcher uses data analysis for interpretation.

Preparation of Report

Research report is a formal document, which has a fixed format.